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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester (CBCS) Degree Examinations, NOV. - DEC. 2025**25CHY101 – CHEMISTRY FOR BIOTECHNOLOGY**

(For BTY)

Duration: 3 Hours]

[Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from each Unit.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- Which of the following is an example for ion selective electrode?
A) Platinum electrode B) Gas electrode
C) Calomel electrode D) Glass electrode
- Emf of the concentration cell $\text{Ag(s)}|\text{Ag}^+(0.01\text{M})||\text{Ag}^+(0.1\text{M})|\text{Ag}$ at 298 °K is
A) 0.0591V B) 0.02955V
C) -0.0591V D) -0.02955V
- Potentiometric titration involves measurement of.....
A) Potential difference between the indicator electrode and reference electrode B) the resistance of an electrolytic solution
C) pH of the solution D) absorbance of the solution
- Compared to chemical analysis, instrumental analysis.....
A) requires very large amount of analyte B) requires no analyte
C) requires very small amount of analyte D) requires no calibration of instruments
- In Li-ion battery, during discharging Li^+ flow through the membrane towards
A) Cathode B) Anode
C) Electrolyte D) None of these
- Which of the following is suitable for powering portable devices such as smartphones and laptops?
A) Lead-acid battery B) Lithium-ion battery
C) Methanol-oxygen fuel cell D) Vanadium flow battery
- Mg water activated (Mg-AgCl) battery is an example forbattery
A) Secondary B) Primary
C) Reserve D) Modern
- Solar cells are the semiconductor devices which converts sunlight into direct current through.....
A) Photoelectric effect. B) Raman effect
C) Sensing property D) Magnetic effect
- The role of EDTA in hardness determination is.....
A) Indicator B) Buffer
C) Oxidizing agent D) Complexing agent
- Hydrolysis of MgSO_4 gives rise to
A) $\text{MgCO}_3 + \text{H}_2\text{SO}_4$ B) $\text{MgCO}_3 + \text{H}_2\text{SO}_4$
C) $\text{MgCO}_3 + \text{HCl}$ D) $\text{Mg(OH)}_2 + \text{H}_2\text{SO}_4$
- Which of the following is TRUE?
A) Lime removes permanent hardness caused due to calcium salts B) Soda removes temporary hardness caused due to calcium salts
C) Lime removes permanent hardness caused due to magnesium salts D) Soda removes permanent hardness caused due to magnesium salts

12. In electro dialysis process, which of the following is TRUE for ion selective membranes?
- A) Cation selective membrane allows the passage of Cl^- ions
B) Cation selective membrane allows the passage of Na^+ ions
C) Anion selective membrane allows the passage of Na^+ ions
D) Ion selective membranes are not used
13. Phenol is monomer
- A) a monofunctional
B) a bifunctional
C) a trifunctional
D) not a monomer
14. Which of the following statement is correct about carbon fibre?
- A) Carbon fibre is a fibre made up of carbon
B) Components made of carbon fibre is used in aerospace and automotive fields
C) Carbon fibre is made by heating polyacrylonitrile
D) All the choices are correct
15. The most conductive form of polyaniline is
- A) Leucoemeraldine
B) Emeraldine salt
C) Pernigraniline
D) All forms are equally conductive
16. The molecular mass (M) of a homopolymer is related to the degree of polymerization (DP) by the equation
- A) $\text{DP} / \text{molecular weight of the repeat unit}$
B) $\text{DP} \times \text{molecular weight of the repeat unit}$
C) $\text{molecular weight of the repeat unit} / \text{DP}$
D) $[\text{DP} \times \text{molecular weight of the repeat unit}] / \text{DP}$
17. Which of the following statements best describes zero waste technology (ZWT)?
- A) It focuses on increasing waste disposal methods.
B) It encourages the production of hazardous waste.
C) It aims to minimize or eliminate waste by reusing materials.
D) It relies solely on recycling methods.
18. The enzymes which catalyzes oxidation-reduction reaction is.....
- A) Oxidoreductase
B) Transferases
C) Hydrolases
D) Ligases
19. In.....approach atoms or molecules are used as building blocks to produce nanomaterials.
- A) Top down
B) Crushing
C) Moulding
D) Bottom up
20. Choose the INCORRECT statement pertaining to Sol-Gel method for nanoparticle synthesis.
- A) Sol is a stable suspension of colloidal particles in a liquid
B) Metal alkoxides are the preferred precursors in sol-gel process
C) Urea is used as fuel.
D) The gel is formed by the polymerization reaction.

PART - B: DESCRIPTIVE ANSWER QUESTIONS

80 Marks

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|--------------|------------|------------|------------|
| 1. a) Derive an expression for electrode potential of glass electrode | 6 | 3 | 1 | 1 |
| b) Explain the theory, principle and application of potentiometry in the ferrous-dichromate titration. | 8 | 2 | 1 | 1 |
| c) Excess potassium thiocyanate solution is added in colorimetric determination of Fe. Justify. | 2 | 2 | 1 | 1 |
| 2. a) Explain the application of calomel electrode for the following: | | | | |
| i) Determination of Zn electrode potential | 8 | 2 | 1 | 1 |
| ii) Determination of pH of an unknown solution | | | | |
| b) Outline the application of conductometry in the estimation of strong acid and acid mixture using strong base. | 8 | 2 | 1 | 1 |

Unit – II

- | | | | | |
|---|---|---|---|---|
| 3. a) Analyze the construction and working of Li-ion battery. | 8 | 3 | 2 | 1 |
|---|---|---|---|---|

b)	Define photovoltaic cells. Give construction, working and applications of PV cells.	8	2	2	1
4. a)	Explain reserve batteries with examples. Mention their uses and advantages	6	2	2	1
b)	Define fuel cell. List any two advantages of fuel cells	3	2	2	1
c)	Outline the construction and working of Vanadium redox flow battery. Mention any two applications of it.	7	2	2	1

Unit – III

5. a)	List the four major types of impurities found in water.	2	1	3	1
b)	Explain Winkler's titration method for the measurement of dissolved oxygen in water.	7	2	3	1
c)	Analyze ion exchange method for the demineralization of water.	7	3	3	1
6. a)	25 mL of an industrial effluent when subjected to 'Chemical Oxygen Demand' test required 22.5 mL of 0.5N $K_2Cr_2O_7$ solution for complete oxidation. Calculate the COD of the sample.	6	3	3	2
b)	Outline the reasons for boiler corrosion and their prevention approaches.	8	2	3	1
c)	Identify any two best methods suitable for the removal of scales in boilers.	2	3	3	1

Unit – IV

7. a)	Classify polymers based on their response to heat and give suitable examples.	4	2	4	1
b)	Explain the mechanism of conduction in polyacetylene through oxidative doping.	7	2	4	1
c)	A coal sample with 93% C, 5% H_2 and 2% ash is subjected to combustion in a bomb calorimeter. Calculate the GCV and LCV of the coal, if mass of coal sample taken is 0.0095kg, mass of water in the copper calorimeter is 2kg, water equivalent of calorimeter is 0.7kg, rise in temperature of water is 2.8K and latent heat of steam is 2457.182 kJ/kg. Specific heat of water = 4.187kJ/kg/°K.	5	3	4	2
8. a)	Analyze the factors influencing glass transition temperature (T_g) value.	6	2	4	1
b)	Define elastomers. Explain the synthesis of silicone rubber and mention any two applications	4	2	4	1
c)	Outline the determination of calorific value of a solid/liquid fuel using bomb calorimeter.	6	2	4	1

Unit – V

9. a)	Outline the major oxide pollutants, their sources, and its effects on humans and environment.	9	2	5	1
b)	Explain the synthesis of carbon nano tubes (CNTs) by Chemical vapour deposition method and mention any two applications of CNTs.	7	2	5	1
10. a)	Explain solvent-free reactions. List out their advantages and disadvantages	6	2	5	1
b)	Illustrate the synthesis of nanomaterials by sol-gel method.	8	2	5	1
c)	Describe catalytic properties of nanomaterials.	2	2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
